Eastern Bering Sea Slope Survey
Gear Description for Poly Nor'Eastern Trawl

Netting: Polyethylene, 5" stretch measure - 4 mm top and sides, 5 mm bottom and intermediate.

Headrope: 89' 1" of ½" (6×19) galvanized wire rope wrapped with ⅜" polypropylene rope.
Both eyes have ½" gusseted thimble; 89' 1" doesn't include the length of either eye.
Headrope length is measure from the top of the nicro sleeve to the top of the nicro at other end.
Top wing is hung over 16.31" per taper combination.
Headrope setback - 18" of ½" long link galvanized alloy chain and connecting hardware.

Bolsh line: 81' 7" plus thimbled eyes of ⅜" (6×19) galvanized wire rope, wrapped with ⅜" polypropylene rope.

Fishing Line: 81' of ½" long link galvanized alloy chain.
Safe working load of 11,300 lbs.

Breastlines: ½" (6×19) galvanized wire rope wrapped with ⅜" polypropylene rope.
Top corner 19' 6", bottom corner 8' 8", bottom side panel 30' 6".
All lengths are measured from the top of nicro sleeve at wing tip and including thimbled eye at ribline.

Riblines: ¾" Samson 2in1® Duralon braided trawl rope.
Riblines are hung at 98% of stretch measure of gored seam.
All measurements are made with 400 lbs of tension on rope.
Gored seams are attached to the ribline using white untreated 60 T braided nylon hanging twine used to tie a benzel every 16".

Groundgear: 8" rubber discs strung on ½" galvanized, long link marine deck lashing chain, grade 70, 79' 6" bearing to bearing (eye to eye). "Flying wing" outboard sections, 4" rubber discs strung on ¾" (6×19) fiber core wire rope.
See groundgear diagram for more detailed description.

Side Seams: Side seams are laced in each panel individually.
Seams are made by gathering 3 meshes (4 knots) and lacing them together using white, double 21 thread perma-grip® (or like kind) nylon, three-strand twine.
Individual panels are laced together using green, double 21 thread perma-grip® nylon, three-strand twine.
Panels which are secured to framing lines have selvage edge created by gathering 3 meshes as described above.
These selvage edges are hung tight to framing lines using 60 T braided nylon hanging twine.

Flotation: Ninety 8" side lug deep-water trawl floats (5.4 lbs of buoyancy each) hung evenly to headrope starting 1' from wing tip, leaving 3' open in the center of the headrope.
Total buoyancy - 486 lbs.

Splitting Gear: ¾" Spectra® rope spliced 21' with eyes in each end.
The rope is passed through four galvanized rings secured to each ribline 18 meshes from the bottom. A second identical splitting strap is located 38 meshes from the bottom.

Codend: 3½" stretched measure (including 1 knot) polyethelene 4 mm double bar mesh.
Four panels cut 30 meshes long by 100 meshes deep.
Two meshes in a gored seam each side, leaving 26 open meshes per panel.

Gored seams are laced together using double 21 T nylon twine.
Riblines in codend are ¾" 2in1® Duralon braided trawl rope, hung at 90% of the stretch measure of the gored seams.
Codend is closed at terminal end using 25 2½" X ¼" galvanized steel rings.
A ¼" Duralon rope is passed through the selvage mesh and a ring is attached to the rope using a cow hitch every 12", with five open meshes between each ring.
The bag is then closed using a ⅝"-¾" hauling clip.

Eastern Bering Sea Slope Survey
Materials List for Poly Nor'Eastern Trawl

Specifications of wire, chain, and rope include rated breaking strength (BS), specific gravity (SG), and density.

WEB: All web in the trawl should be depth stretched and heat set in the manufacturing process. Mesh sizes listed in the plan are given in "stretched measure", a standard method of measuring mesh size that includes the length of one knot.
Top, body and wings, side panels ~ 5", 4-mm polyethylene knotted web.
Bottom, body and wings, intermediate ~ 5", 5mm polyethylene knotted web. All polyethylene web in net panels is orange.
Liner ~ 1.25", # 18T three strand nylon knotted web - dyed green.
Chaffing gear ~ 6", 6mm double bar mesh, polyethylene knotted web - orange.
Chaffing strip, bottom~ 5", 5mm double bar mesh, polyethylene knotted web - orange.

WIRE ROPE: ½" 6×19 galvanized fiber core (BS 9,700 kg, 0.62 kg/m), eyes are formed using galvanized reinforced thimbles.

CHAIN: ½" (16 mm) galvanized, long link marine deck lashing chain, grade 70 (BS 20,500 kg, 3.42 kg/m).

FLOATS: Ninety 8" side lug deep-water trawl floats (5.4 lbs buoyancy each) depth rated to a minimum of 2,000 m.

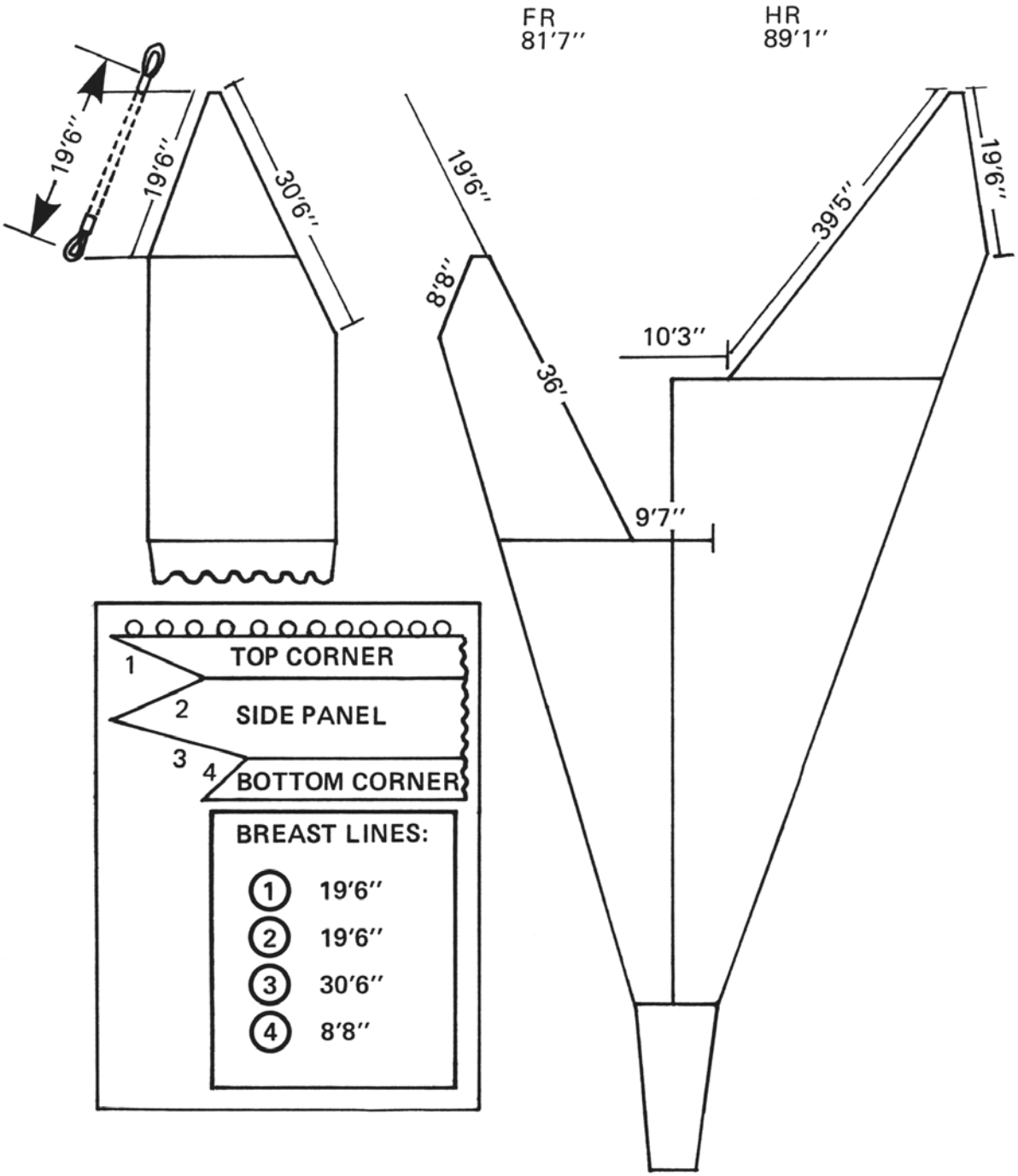
RIBLINES: ¾" Samson Duralon™ 2-in-1® stable braid, white, light bonding (BS 8,800 kg, SG 1.83, 0.27 kg/m).

TWINE: Sewing panels together: 5-mm polyethylene twine, any color other than orange.
Lacing seams: # 21T white, three strand nylon.
Joining top and side seams together: # 21T green three strand nylon.
Joining bottom and intermediate seams: 60T white braided hanging twine.

ROPE: Float lines - 1/4" polypropylene (BS 570 kg, SG 0.95).
Pucker ring lines- Samson 1/4" Duralon™ 2-in-1®(BS 1,270 kg, SG 1.38)
Restrictor rope- 1 1/8" Poly Plus™ three strand (BS 10,740 kg, SG 0.91).
Splitting strap- 3/4" Spectra™ braided rope with eye each end (BS 21,773 kg, SG 0.98).
Rope to serve cables- 3/8" polypropylene three strand.
Chaffing gear- Poly hula skirt material.

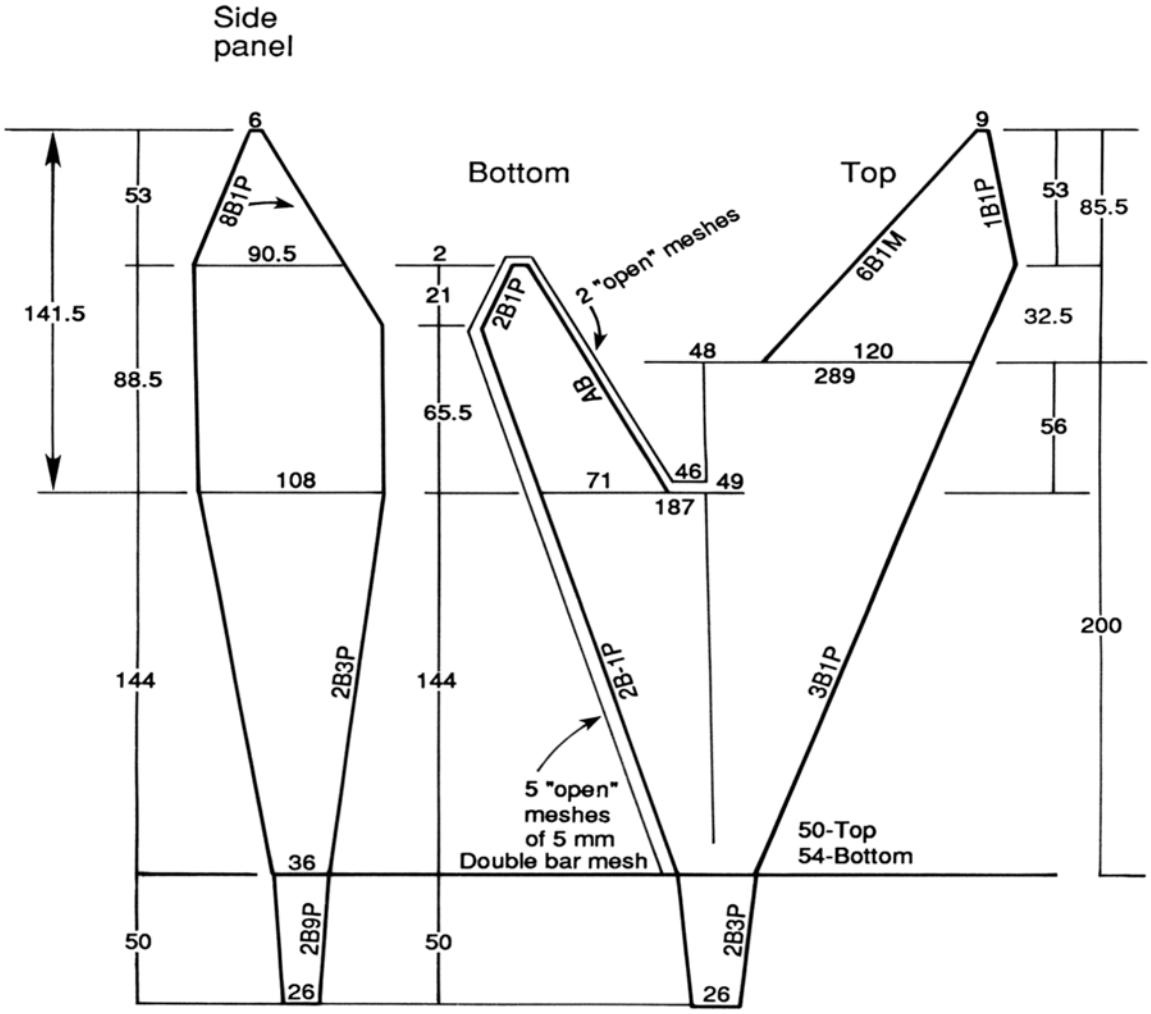
FOOTROPE: ½" galvanized, long link marine deck lashing chain, grade 70, 79' 6".
8" rubber disc with a 2½" center hole.
Steel plate chain washers- 5" x ⅜" - 2½" center hole.
Chain toggles- 5.5 lbs. each, chain droppers are 6½" long, spaced 18" on center along length of footrope. Toggles are attached to fishing line with 5/16" import trawl shackles. Four hole triangle delta plate, ⅜" steel.

Eastern Bering Sea Slope Bottom Trawl Survey
Framing Lines for Poly Nor'Eastern Trawl



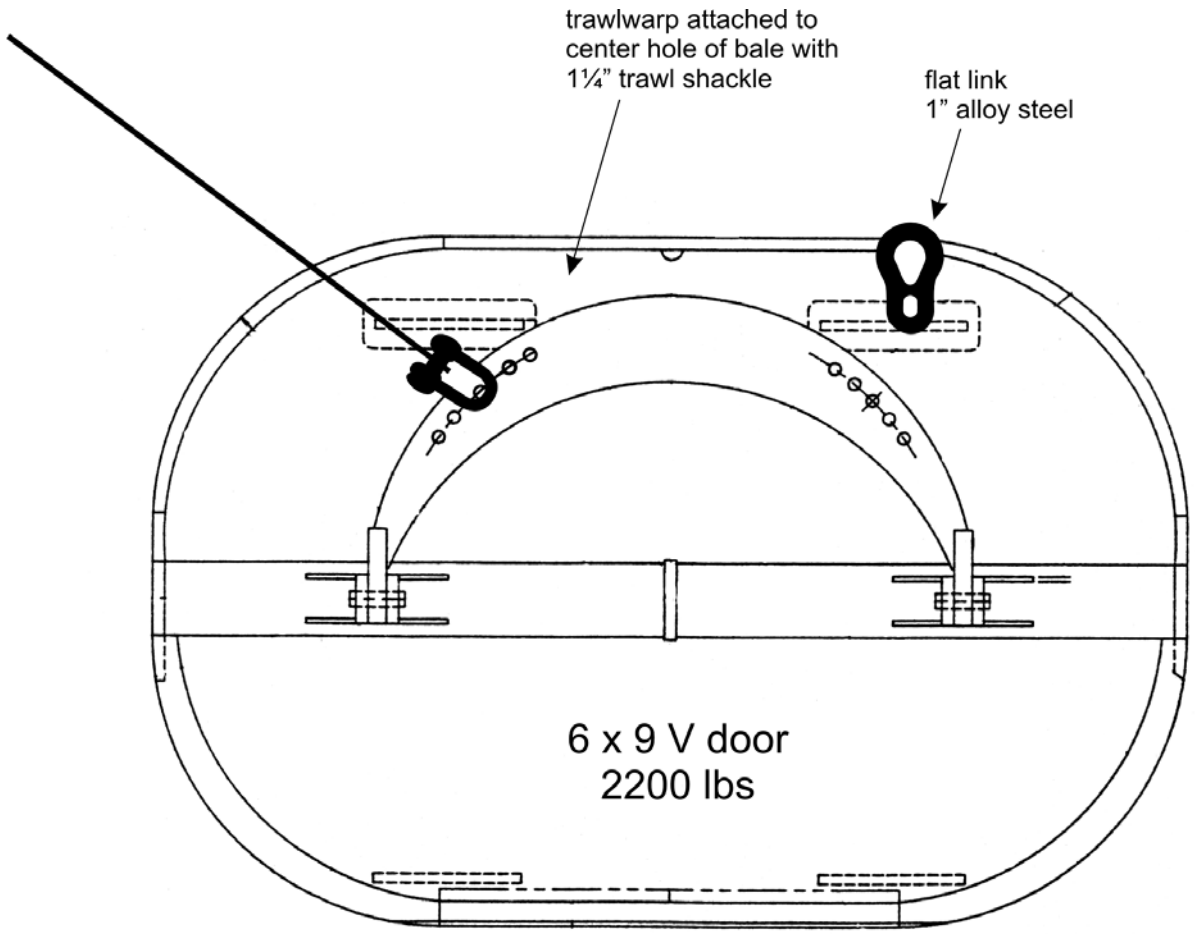
Eastern Bering Sea Slope Bottom Trawl Survey
Net Plan for Poly Nor'Eastern Trawl
("Cut Plan" = Total Mesh Counts)

Twine Sizes: top and sides 4 mm
bottom and intermediate 5 mm

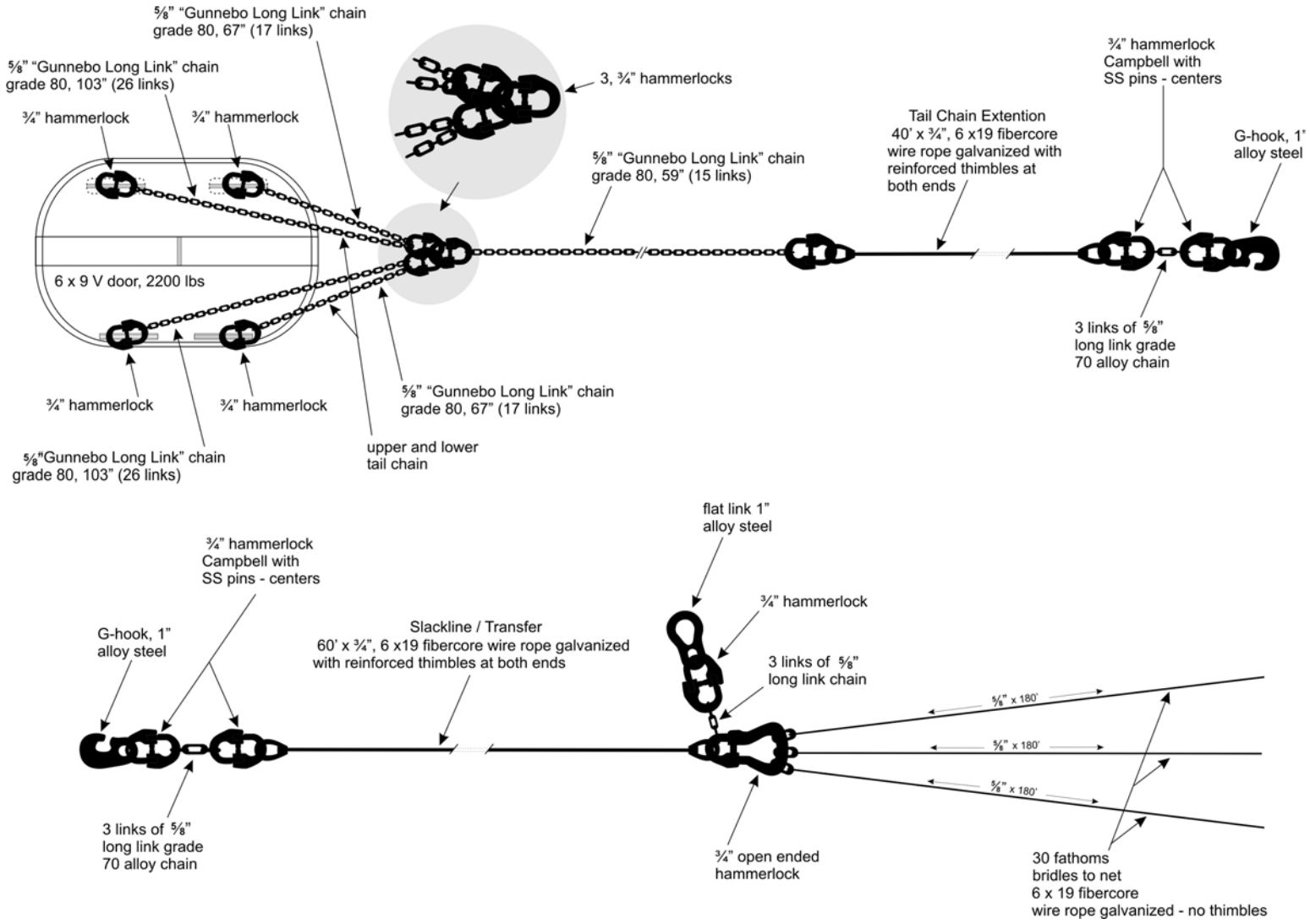


Web: Chaffing strip along inside of Bottom wings and Busom. Cut 8 meshes wide. 5 mm Double Bar mesh, goring 3 meshes on each side (leaving 2 open meshes). Secure 3 mesh of gore on inside (Bar Cut) of Bottom wings, and securing other gore to footrope (Bolsh).

Eastern Bering Sea Slope Bottom Trawl Survey
Trawl Door Rigging Plan Detail for Poly Nor'Eastern Trawl
Sole Manufacturer NET Systems, Inc., Bainbridge Island, WA

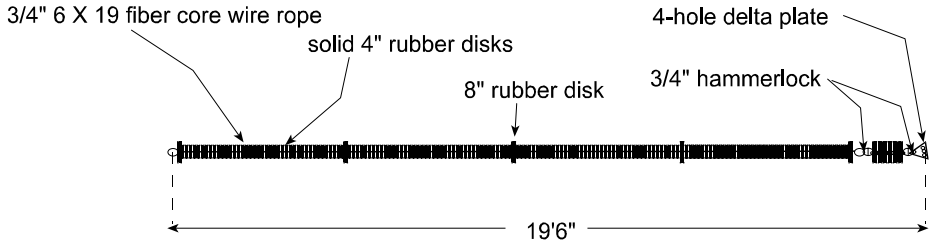


Eastern Bering Sea Slope Bottom Trawl Survey
Trawl Door Rigging Plan for Poly Nor'Eastern Trawl

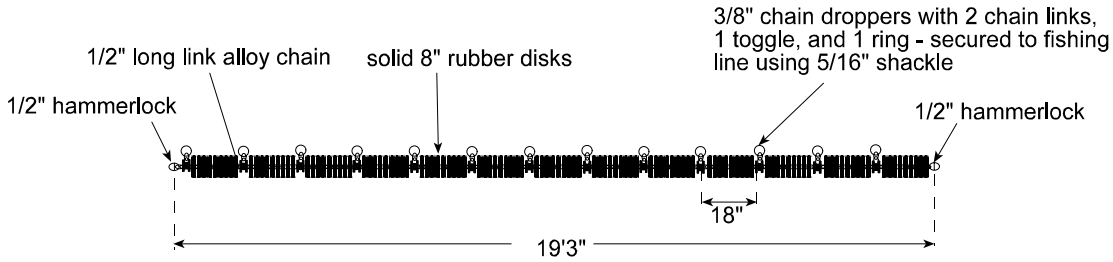


Eastern Bering Sea Slope Bottom Trawl Survey
Groundgear Construction Plan for Poly Nor'Eastern Trawl

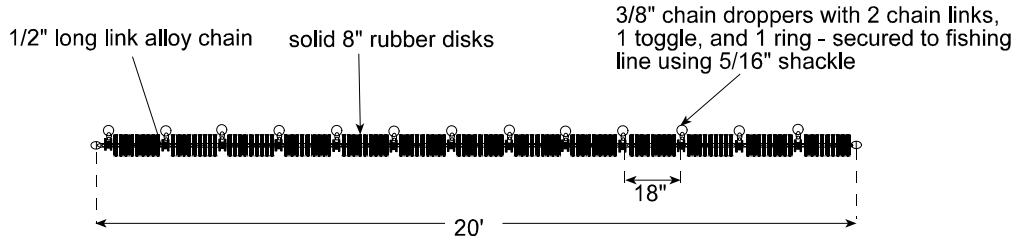
Outboard section



Middle section

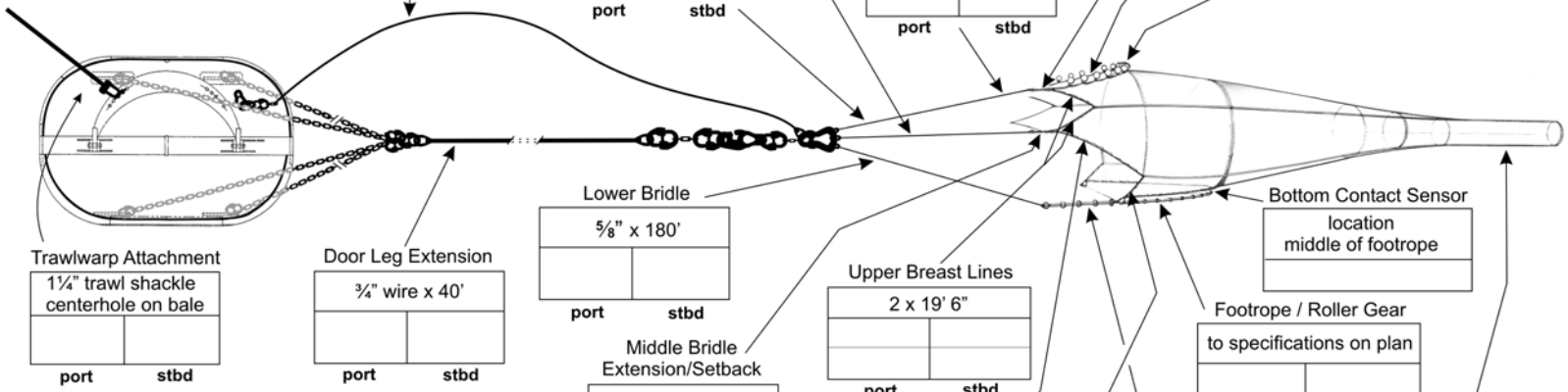


Inboard section



Vessel	
Cruise	
Date	
Checked by	
Checked by	
Net I.D. number	

Survey Trawl Check List
Eastern Bering Sea Slope Bottom Trawl Survey
Poly Nor'Eastern Trawl



Trawl Door #
6 x 9 V door
2200 lbs
port stbd

Slackline/Transfer
¾" wire x 60'
port stbd

Middle Bridle
⅝" x 180'
port stbd

Upper Bridle Extension/Setback
½" LL chain x 18"
port stbd

Headrope
89' 1"

Floats
8" x deep water x 90 / 45 on ea. side
port stbd

Upper Bridle
⅝" x 180'
port stbd

Scanmar / Netmind wire stop location
Upper bridle 12" from nico press
port stbd

Trawlwarp Attachment
1¼" trawl shackle centerhole on bale
port stbd

Door Leg Extension
¾" wire x 40'
port stbd

Lower Bridle
⅝" x 180'
port stbd

Middle Bridle Extension/Setback
½" LL chain x 9"
port stbd

Upper Breast Lines
2 x 19' 6"
port stbd

Middle Breast Line
30' 6"
port stbd

Bottom Breast Line
8' 8"
port stbd

Bottom Contact Sensor
location middle of footrope

Footrope / Roller Gear
to specifications on plan
port stbd

Flying Wing
¾" wire x 19'6" w/disks
port stbd
(Measure from bearing point on wire to fishing line attachment point on delta plate)

Codend	
Liner	
Splitting strap	
Cowbell	

4 Part Door Leg

	⅝" LL chain			door hanging angle 35°- 40°
	front 103"		back 67"	single 59"
port				
stbd				